MATH 5321 Exam II 1 April 2009

Make-Up

Answer the problems on separate paper. You do <u>not</u> need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **<u>Show all relevant supporting steps!</u>**

1. X

2. X

3. (20 pts) Find the number of zeros of $p(z) = z^6 - 6z^4 - 10z - 2$

   a.) in $B(0,1)$

   b.) in $B(0,2)$

   c.) in $B(0,3)$

4. (20 pts) Let G be a bounded region in $\mathbb{C}$. Let $\{f_n\} \subset \mathcal{C}(\overline{G}, \mathbb{C}) \cap \mathcal{A}(G)$ and let $f \in \mathcal{C}(\overline{G}, \mathbb{C}) \cap \mathcal{A}(G)$. Suppose that $f_n \to f$ uniformly on ∂G. Show that $f_n \to f$ in $\mathcal{C}(G, \mathbb{C})$.